

$$d. \begin{cases} 2x + 3y = 6 \\ x - y = 1 \end{cases}$$

$$\text{Soln} \quad |A| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = 2 \times (-1) - 1 \times 3 = -2 - 3 = -5$$

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{-5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$= \frac{-1}{5} \begin{bmatrix} -1 \times 6 - 3 \times 1 \\ -1 \times 6 + 2 \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5/5 \\ 1/5 \end{bmatrix}$$

$$e. \begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$$

$$\text{Soln:} \quad |A| = \begin{vmatrix} 4 & 3 \\ 1 & -3 \end{vmatrix} = -12 - 3 = -15$$

$$A^{-1} = \frac{1}{-15} \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{-15} \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ -1 \end{bmatrix}$$

$$= \frac{1}{-15} \begin{bmatrix} -3 \times 11 + (-3) \times (-1) \\ -1 \times 11 + 4 \times (-1) \end{bmatrix} = \begin{bmatrix} -33 + 3 \\ -11 - 4 \end{bmatrix} = \begin{bmatrix} -30 \\ -15 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} \therefore x=2, y=1$$

f. $2x + 3y = 8$
 $4x + 6y = 20$

Solⁿ: $|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0 \therefore \text{No sol}^n$

Qo. $x + y - z = 2$
 $x + y + z = 7$
 $2x + y + z = 13$

Solⁿ: $x = 7 - z$
(i) $7 - z + y - z = 2$
or, $7 - 2z + y = 2$
or, $5 + y = 2z$
 $\therefore z = \frac{5+y}{2}$

Now

(iii) $2(7-z) + y + \frac{5+y}{2} = 13$

or, $7 + 3y - 10 - 2z = 0$

or, $-3 + y = 0$

$\therefore y = 3$

or, $14 - 2z + y + \frac{5+y}{2} = 13$

$\therefore z = \frac{5+3}{2} = \frac{8}{2} = 4$

or, $28 - 4z + 2y + 5 + y = 26$

$\therefore x = 7 - z = 7 - 4 = 3$

or, $7 + 3y - 4z = 0$

$\therefore x = 3, y = 3, z = 4$

or, $7 + 3y - \frac{4(5+y)}{2} = 0$

$$(ii) \quad 2x(-3z) - 2 + 6z = 0$$

$$or, \quad -6z - 2 = 6z$$

$$\therefore y = 2x1 = 2$$

$$or, \quad x - 3x1 = -1 \quad \therefore x = -1$$

$$d. \quad x + 4y - z = 4$$

$$2x + 5y + 8z = 15$$

$$x + 3y - 3z = 1$$

Soln:

Subtracting eqⁿ (iii) from (i)

$$(x + 4y - z) - (x + 3y - 3z) = 4 - 1$$

$$\therefore y + 2z = 3$$

Subtract eqⁿ (iii) from (ii)

$$(2x + 5y + 8z) - (x + 3y - 3z) = 15 - 1$$

$$\therefore x + 2y + 11z = 14$$

$$Now, \quad y = 3 - 2z$$

$$x + 2(3 - 2z) + 11z = 14$$

$$or, \quad x + 6 - 4z + 11z = 14$$

$$\therefore x + 7z = 8$$

$$Now, \quad x + 3(3 - 2z) - 3z = 1$$

$$or, \quad x + 9 - 6z - 3z = 1$$

$$or, \quad x - 9z = -8$$

$$From, \quad x + 7z = 8$$

$$x - 9z = -8$$

$$or, \quad 16z = 16 \quad \therefore z = 1$$

$$\therefore y = 3 - 2 \times 1 = 1$$

$$\therefore x + 7 \times 1 = 8 \Rightarrow x = -1$$

e. $5x + 3y + 9z = -1$
 $-2x + 3y - 2 = -2$
 $-x - 4y + 5z = 1$

Solⁿ: Sub eqⁿ (ii) from (i)
 $(5x + 3y + 9z) - (-2x + 3y - 2) = -1 - (-2)$
 $\therefore 7x + 10z = 1$

Now, Add eqⁿ (ii) & (iii)
 $-2x + 3y - 2 + -x - 4y + 5z = -2 + 1$

$\therefore -3x - y + 4z = -1$
 From, solving, $z = 0$

So,
 Eqⁿ (i), $5x + 3y = -1$
 (ii) $-2x + 3y = -2$
 Subtract, $7x = 1$
 $x = 1/7$

f. $6x - 5y + 2z = 3$
 $2x + y - 4z = 5$
 $3x - 3y + 2z = -1$

Solⁿ: Multiply eqⁿ (ii) by 3
 $6x + 3y - 12z = 15$

Now, Subtract eqn (i)
 $8y - 14z = 12$

Now, Multiply (iii) by 2
 $6x - 6y + 2z = -2$

Now, Subtract eqn (i), $y = 5$
 Using eqn (ii), $2x + 5 - 4z = 5$
 or, $2x - 4z = 0$
 or, $x = 2z$

3 $f(x) = \sqrt{x} \cos x$

let $u = \sqrt{x} = x^{1/2}$, $v = \cos x$

Now, $\frac{du}{dx} = \frac{d}{dx} (x^{1/2}) = \frac{1}{2\sqrt{x}}$

$\frac{dv}{dx} = \frac{d}{dx} (\cos x) = -\sin x$

$\therefore f'(x) = \frac{1}{2\sqrt{x}} \cos x - \sqrt{x} \sin x$

4. $f(x) = x^3 e^x$

let $u = x^3$, $v = e^x$

$\frac{du}{dx} = \frac{d}{dx} x^3 = 3x^2$

$\therefore f'(x) = 3x^2 e^x - x^3 e^x = e^x (3x^2 - x^3)$

5 $f(x) = \frac{x^3}{e^x}$

or, $f(x) = x^3 e^{-x}$

let $u = x^3$, $v = e^{-x}$

$\frac{du}{dx} = 3x^2$, $\frac{dv}{dx} = -e^{-x}$

$\therefore f'(x) = 3x^2 e^{-x} - x^3 e^{-x}$

6 $f(x) = (3x^2 + 2x + 1)^5$

let $u = 3x^2 + 2x + 1$

so, $f(x) = u^5$

$\frac{du}{dx} = \frac{d}{dx} (3x^2 + 2x + 1) = 6x + 2$

$u' = 6x + 2$

$$\therefore f'(x) = 5(3x^2 + 2x + 1)^4 (6x + 2)$$

7 $f(x) = (\sin x)^4$

let $u = \sin x$

Then, $f(x) = u^4$

$$\frac{d}{dx} u = \frac{d}{dx} (u^4) = 4u^3 \cdot u'$$

$$u' = \sin x \cos x$$

$$\therefore f'(x) = 4 \sin^3(x) \cos x$$

8 $f(x) = e^{(3x^2 + 2x)}$

let $u = 3x^2 + 2x$

Then, $f(x) = e^u$

$$\text{So, } \frac{d}{dx} u = \frac{d}{dx} (e^u) = e^u \cdot u'$$

$$u' = 6x + 2$$

$$\therefore f'(x) = e^{3x^2 + 2x} (6x + 2)$$

9 $f(x) = \ln(5x^2 + 3x)$

let $u = 5x^2 + 3x$

$f(x) = \ln u$

$$\text{So, } \frac{d}{dx} u = \frac{d}{dx} (\ln u) = \frac{u'}{u}$$

$$u' = 10x + 3$$

$$\therefore f'(x) = \frac{10x + 3}{5x^2 + 3x}$$

10. $f(x) = \sqrt{x^3 + 4x + 1}$
 $f(x) = (x^3 + 4x + 1)^{1/2}$

let,

$$u = x^3 + 4x + 1$$

Then, $f(x) = u^{1/2}$

$$f'(x) = \frac{1}{2} u^{-1/2} u'$$

$$u' = 3x^2 + 4$$

$$f'(x) = \frac{1}{2} (x^3 + 4x + 1)^{-1/2} (3x^2 + 4)$$

$$\therefore f'(x) = \frac{3x^2 + 4}{2\sqrt{x^3 + 4x + 1}}$$

Problem 2

Given: $g(x) = x^2 - 3x + 2$

i) Derivative of $f(x) \cdot g(x)$ at $x=1$

$$g(1) = 1^2 - 3 \times 1 + 2 = 0$$

$$g'(x) = 2x - 3$$

Chain rule: $(fg)'(1) = f'(1)g(1) + f(1)g'(1)$
 $= 0 \times 0 + 5 \times (-1)$
 $= -5$

As, $f(0) = 3$ & $f'(0) = -1$

$$g(0) = 0^2 - 3 \times 0 + 2 = 2$$

$$g'(0) = 2 \times 0 - 3 = -3$$

Now, $3(2z) - 3 \times 5 + z = -1$

or, $6z - 15 + z = -1$

or, $7z = 14$

$\therefore z = 2$

$\therefore x = 4$

$\therefore y = 5$

Q.1.1) $f(x) = x + \sqrt{x}$

$f'(x) = \frac{d}{dx} (x + \sqrt{x})$

$= \frac{dx}{dx} + \frac{d\sqrt{x}}{dx}$

$= 1 + \frac{d}{dx} (x^{1/2})$

$= 1 + \frac{1}{2} x^{1/2-1}$

$= 1 + \frac{1}{2} x^{-1/2}$

$= 1 + \frac{1}{2\sqrt{x}}$ //

2. $f(x) = 2 + \frac{1}{x} + \frac{1}{x^2}$

or, $f(x) = 2 + x^{-1} + x^{-2}$

$f'(x) = \frac{d}{dx} (2) + \frac{d}{dx} (x^{-1}) + \frac{d}{dx} (x^{-2})$

$= 0 + -x^{-2} - 2x^{-3}$

$= -\frac{1}{x^2} - \frac{2}{x^3}$ //

$$h(-2) = \frac{4}{2} + 4 = \frac{4+8}{2} = \frac{12}{2} = 6$$

$$h(3) = \frac{9}{2} + 16 = \frac{9-12}{2} = -\frac{3}{2}$$

$$h(2) = \frac{4}{2} - 4 = \frac{4-8}{2} = \frac{-4}{2} = -2$$

$\therefore$ Min at $h(2) = -2$ & max. at $h(-2)$.

3 $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2]$

$$p'(x) = 9x^2 - 18x$$

$$p(-1) = -3 - 9 + 7 = -5$$

$$p(2) = 3 \times 8 - 9 \times 4 + 7 = 24 - 36 + 7 = -5$$

$$p(0) = 7$$

$\therefore$ Min. at $p(-1)$ or $p(2)$ & max. at $p(0)$

4 $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

$$f'(x) = 2x - 4$$

$$f(0) = 0 - 0 + 6 = 6$$

$$f(3) = 9 - 12 + 6 = 9 - 6 = 3$$

Min. at $f(3)$ & max. at $f(0)$

5 $q(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$

$$q'(x) = -6x^2 + 6x + 12$$

$$q(-1) = 2 + 3 - 12 = -7$$

$$q(2) = -16 + 12 + 24 = -16 + 36 = 20$$

$\therefore$ Min. at $q(-1)$ & max. at $q(2)$

$$\begin{aligned} \text{c. } -x - 2y + z &= 1 \\ 2x + 3y &= 2 \\ y - 2z &= 0 \end{aligned}$$

Soln: $y = 2z$

(ii) $2x + 3y = 2$

or, $2x + 3 \times 2z = 2$

or, $2x + 6z = 2$

or, $2x - 2 + 6z = 0$ Divide by 2, $x + 3z = 1$

Now

(i) $-x - 2y + z = 1$

or, $-2y + z = x$

or, $-2 \times 2z + z = x$

or, $-4z + z = x$

$\therefore x + 3z = -1$

From, $x + 3z = 1$

$x + 3z = -1$

$\therefore 2 = 1$

4.3.3

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1. $f(x,y) = x^2 + y^2$ at:

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = 2y$$

$$\nabla f = (2x, 2y)$$

a. $(0,0) = (2 \times 0, 2 \times 0) = (0,0)$

b. $(1,3) = (2 \times 1, 2 \times 3) = (2,6)$

c. $(-1,-5) = (2 \times -1, 2 \times -5) = (-2,-10)$

2. $f(x,y,z) = x^3z - 2y^2x + 5z$

$$\frac{\partial f}{\partial x} = 3x^2z - 2y^2$$

$$\frac{\partial f}{\partial y} = -4yx$$

$$\frac{\partial f}{\partial z} = x^3 + 5$$

$$\nabla f = (3x^2z - 2y^2, -4yx, x^3 + 5)$$

a. $(-1,1,-4) = (3 \times 1 \times (-4) - 2 \times 1, -4 \times 1 \times 1, 1 + 5) = (-14, -4, 6)$

b. $(0,1,0) = (3 \times 0 \times 0 - 2 \times 1, -4 \times 1 \times 0, 0 + 5) = (-2, 0, 5)$

c. $(-3,-2,1) = (3 \times 9 \times 1 - 2 \times 4, -4 \times (-2) \times (-3), (-3)^3 + 5) = (19, -24, -22)$

3. $m(x,y) = \ln(x^2 + y^2 + 1)$

$$\frac{\partial m}{\partial x} = \frac{2x}{x^2 + y^2 + 1}$$

$$\frac{\partial m}{\partial y} = \frac{2y}{x^2 + y^2 + 1}$$

$$\nabla m = \left(\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1} \right)$$

a) at $(0,0) = \left(\frac{2 \times 0}{0 + 0 + 1}, \frac{2 \times 0}{0 + 0 + 1} \right) = (0,0)$

b. at $(1, -1) = \left(\frac{2x}{1^2 + 1^2 + 1}, \frac{2x(-1)}{1+1+1} \right) = \left(\frac{2}{3}, -\frac{2}{3} \right)$

c. at $(-2, 3) = \left(\frac{2x(-2)}{4+9+1}, \frac{2x3}{4+9+1} \right) = \left(-\frac{2}{7}, \frac{3}{7} \right)$

2. Maxima & Minima of Multivariate function

1. $m(x, y) = -x^2 - y^2 + 4x + 6y$

$$\frac{\partial m}{\partial x} = -2x + 4 = 0 \Rightarrow x = 2$$

$$\frac{\partial m}{\partial y} = -2y + 6 = 0 \Rightarrow y = 3$$

$$m(2, 3) = -4 - 9 + 8 + 18 = 13$$

$$m(0, 0) = 0, \quad m(4, 0) = 0, \quad m(0, 5) = 5, \quad m(4, 5) = 5$$

$$m(2, 0) = 4, \quad m(2, 5) = 5, \quad m(0, 3) = 9, \quad m(4, 3) = 9$$

$\therefore$ Max at $(2, 3)$ & Min. at $(0, 0)$ & $(4, 0)$.

2. $g(x, y) = xy + x^2 - 4y + 6$

$$\frac{\partial g}{\partial x} = y + 2x = 0$$

$$\frac{\partial g}{\partial y} = x - 4 = 0 \Rightarrow x = 4 \text{ (Outside interval)}$$

$$g(-2, 3) = -6 + 4 - 12 + 6 = -8$$

② Derivative of $f(x)$ at $x=0$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$\left(\frac{f}{g}\right)'(0) = \frac{(-1) \times 2 - 3 \times (-3)}{2^2} = \frac{-2+9}{4} = \frac{7}{4}$$

③ Derivative of $f(g(x))$ at $x=0$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\therefore g(0) = 2$$

$$\text{As, } f'(2) = 3 \text{ and } g'(x) = 2x - 3$$

$$g'(0) = -3$$

$$\text{Substituting, } [f(g(x))]'_{x=0} = f'(2) \cdot (-3) \\ = 3(-3) \\ = -9$$

4.2.1) $g(x) = x^2 - 4x + 5$ at interval $[1, 4]$

$$g'(x) = 2x - 4$$

$$g'(1) = 2 \times 1 - 4 = 2 - 4 = -2 \quad \therefore g'(4) = 0$$

$$g(1) = 1 - 4 + 5 = 2$$

$$g(4) = 16 - 16 + 5 = 5$$

$\therefore$ Min at $g(1)$ & max at $g(4)$

1) $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$

$$h'(x) = x - 2$$

$\therefore x = 2$ so, $x = 2$ is critical point

5 $f(x, y) = x^2 + y^2 - 4x - 6y + 13$

$$\frac{\partial f}{\partial x} = 2x - 4 = 0 \rightarrow x = 2$$

$$\frac{\partial f}{\partial y} = 2y - 6 = 0 \rightarrow y = 3$$

$$f(2, 3) = 4 + 9 - 8 - 18 + 13 = 0$$

$$f(1, 1) = 1 + 1 - 4 - 6 + 13 = 5$$

$$f(3, 1) = 9 + 1 - 12 - 6 + 13 = 5$$

$$f(1, 4) = 1 + 16 - 4 - 24 + 13 = 2$$

$$f(3, 4) = 9 + 16 - 12 - 24 + 13 = 2$$

$\therefore$ Max. at $f(1, 1)$ & $f(3, 1)$

$\therefore$ Min. at $(2, 3)$

$$g(2,1) = 2+4-4+6=8$$

$$g(2,3) = 6+4-12+6=4$$

On edge $x=-2$: $g = -2y+4-4y+16 = 10-6y \rightarrow \text{max at } y=1: g=4$
 $\text{min at } y=3: g=8$

$$g(-2,4) = -8+64-16+6 = 46$$

3. $p(x,y) = x^3 + 3xy + 4$ interval $x: [0,2]$ and $y: [0,3]$

$$\frac{\partial p}{\partial x} = 3x^2 + 3y = 0 \Rightarrow y = -x^2$$

$$\frac{\partial p}{\partial y} = 3x = 0 \Rightarrow x = 0 \text{ then, } y = 0$$

$$p(0,0) = 4$$

$$p(2,0) = 8+4=12$$

$$p(0,3) = 4$$

$$p(2,3) = 8-18+4 = -6$$

$\therefore$ Max. at $p(2,0)$ & min. at $p(2,3)$

4. $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$

$$\frac{\partial h}{\partial x} = 4x - 12 = 0 \Rightarrow x = 3$$

$$\frac{\partial h}{\partial y} = 6y - 18 = 0 \Rightarrow y = 3$$

$$h(3,3) = 18+27-36-54+20 = -25$$

$$h(5,0) = 2 \times 25 + 0 - 12 \times 5 - 0 + 20 = 10$$

$$h(1,0) = 2+0-12-0+20 = 10$$

$$h(1,4) = 2 \times 1 + 3 \times 16 - 12 \times 1 - 18 \times 4 + 20 = -14$$

$$h(5,4) = 2 \times 25 + 3 \times 16 - 12 \times 5 - 18 \times 4 + 20 = -14$$

$\therefore$ Max. at $(1,0)$ & $(5,0)$, min. at $(3,3)$

2. Compute AD and DA

Given

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$O = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$AD = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 2 + 1 \times 0 + 1 \times 0 & 1 \times 0 + 1 \times 3 + 1 \times 0 & 1 \times 0 + 1 \times 0 + 1 \times 4 \\ 1 \times 2 + 2 \times 0 + 3 \times 0 & 1 \times 0 + 2 \times 3 + 3 \times 0 & 1 \times 0 + 2 \times 0 + 3 \times 4 \\ 1 \times 2 + 3 \times 0 + 4 \times 0 & 1 \times 0 + 3 \times 3 + 4 \times 0 & 1 \times 0 + 3 \times 0 + 4 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

$$DA = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 1 + 0 \times 1 + 0 \times 1 & 2 \times 1 + 0 \times 2 + 0 \times 3 & 2 \times 1 + 0 \times 3 + 0 \times 4 \\ 0 \times 1 + 3 \times 1 + 0 \times 1 & 0 \times 1 + 3 \times 2 + 0 \times 3 & 0 \times 1 + 3 \times 3 + 0 \times 4 \\ 0 \times 1 + 0 \times 1 + 4 \times 1 & 0 \times 1 + 0 \times 2 + 4 \times 3 & 0 \times 1 + 0 \times 3 + 4 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

$\therefore AD \neq DA$

3. Verify $AB=AC$

Given, $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 1 + 2 \times 3 \\ 2 \times 2 + 4 \times 1 & 2 \times 1 + 4 \times 3 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 \times 4 + 2 \times 0 & 1 \times 3 + 2 \times 2 \\ 2 \times 4 + 4 \times 0 & 2 \times 3 + 4 \times 2 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$\therefore AB=AC$ //

4. Prove: $(AB)^T = B^T A^T$

let:

A be an $m \times n$ matrix

B be an $n \times p$ matrix

Then, the product $C=AB$ is an $m \times p$ matrix.

The element of C is $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$

Taking transpose of AB ,
 $(AB)^T_{ji} = AB_{ji}$

Substituting the formula of matrix multiplication:

$$(AB)_{ji} = \sum_{k=1}^n a_{jk} \cdot b_{ki}$$

Now, $A^T_{kj} = a_{jk}$
 $B^T_{ik} = b_{ki}$

After multiplying & substituting, $\sum_{k=1}^n a_{jk} b_{ki}$

1.4) Given sets :

a. $A = \{x : x \in \mathbb{N} \text{ and } x \text{ is even}\}$
→ Set of even natural numbers $A = \{2, 4, 6, 8, 10, 12, 14, \dots\}$

b. $B = \{x : x \in \mathbb{N} \text{ and } x \text{ is prime}\}$
→ Set of prime numbers $B = \{2, 3, 5, 7, 11, 13, 17, \dots\}$

c. $C = \{x : x \in \mathbb{N} \text{ and } x \text{ is multiple of } 5\}$
→ Multiple of 5 $C = \{5, 10, 15, 20, 25, \dots\}$

a) $A \cap B = \{2\}$

b) $B \cap C = \{5\}$

c) $A \cup B = \{2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 17, \dots\}$

d) $A \cap (B \cup C) = \{2, 10, 20, 30, \dots\}$

Matrix Operation

1a) 3×3 matrix with ones on the diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) Three-by-four matrix with ones on the diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

c) 4×3 matrix with ones on the diagonal

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

comparing $(AB)_{ij}^T = \sum_k a_{jk} b_{ki}$
 $(B^T A^T)_{ij} = \sum_k a_{jk} b_{ki}$

$\therefore (AB)^T = B^T A^T$

5. let $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

Soln: $A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$A^T \cdot A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

$= \begin{bmatrix} a^2 + b^2 + c^2 & ad + be + cf \\ ad + be + cf & d^2 + e^2 + f^2 \end{bmatrix}$

A matrix is symmetric if $M = M^T$

Above, element (1,2) and (2,1) have $ad + be + cf$

So, $A^T A = (A^T A)^T$

So, $A^T A$ is symmetric

6. Find inverse of matrix

Soln: $A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$

$|A| = \begin{vmatrix} 5 & 6 \\ 4 & 5 \end{vmatrix} = 5 \times 5 - 4 \times 6 = 25 - 24 = 1$

$A^{-1} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$

$$|B| = \begin{vmatrix} 6 & 4 \\ 3 & 3 \end{vmatrix} = 6 \times 3 - 3 \times 4 = 18 - 12 = 6$$

$$B^{-1} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

7. Verify A and B are inverses
Given,

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 4 \times 1 + 1 \times (-3) & 4 \times (-1) + 1 \times 4 \\ 3 \times 1 + 1 \times (-3) & 3 \times (-1) + 1 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 \times 4 - 1 \times 3 & 1 \times 1 - 1 \times 1 \\ -3 \times 4 + 4 \times 3 & -3 \times 1 + 4 \times 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore AB = BA = I$$

8. Verify matrices are inverses

Given,

$$AB = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$